

# Robot Learning

## Project

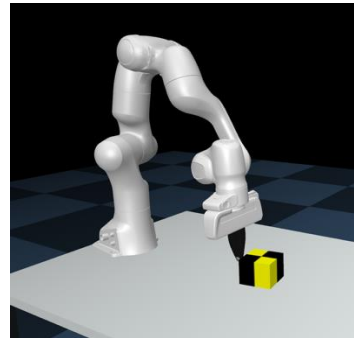
## Topics of today

- Sim-to-real Transfer and Domain Randomization
- Hopper environment
- Uniform Domain Randomization
- Project extension

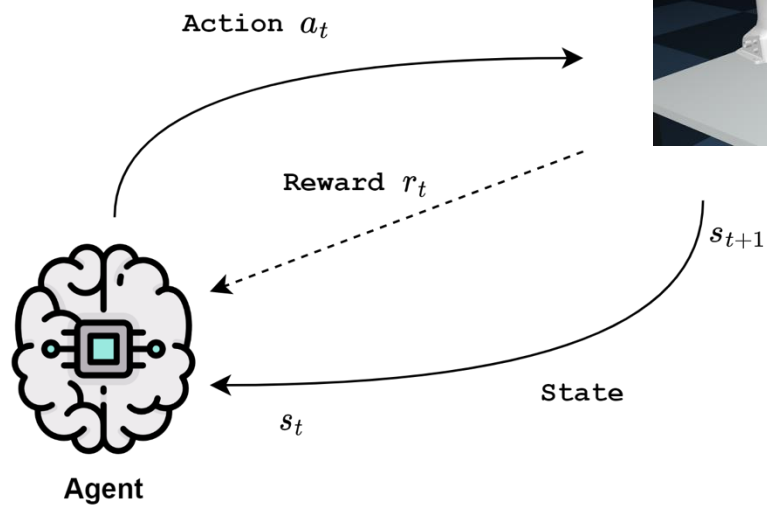
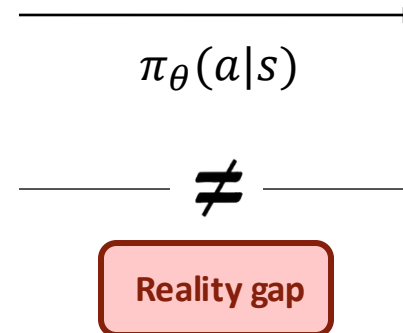
# Sim-to-Real Transfer and Domain Randomization

**Problem:** teach a robot how to push a box to a target location

**Simulation (Source)**  
Training in simulation



**Real World (Target)**  
Testing in the real world

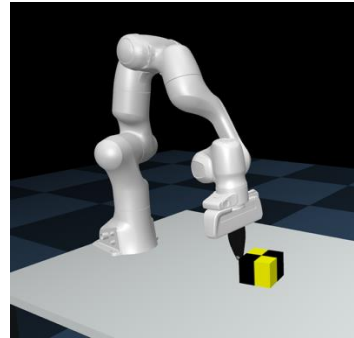


# Sim-to-Real Transfer and Domain Randomization

**Problem:** teach a robot how to push a box to a target location

## Simulation (Source)

Training in simulation on **randomized dynamics**



$$\xi \sim p_{\phi}(\cdot)$$

$$\pi_{\theta}(a|s)$$

## Real World (Target)

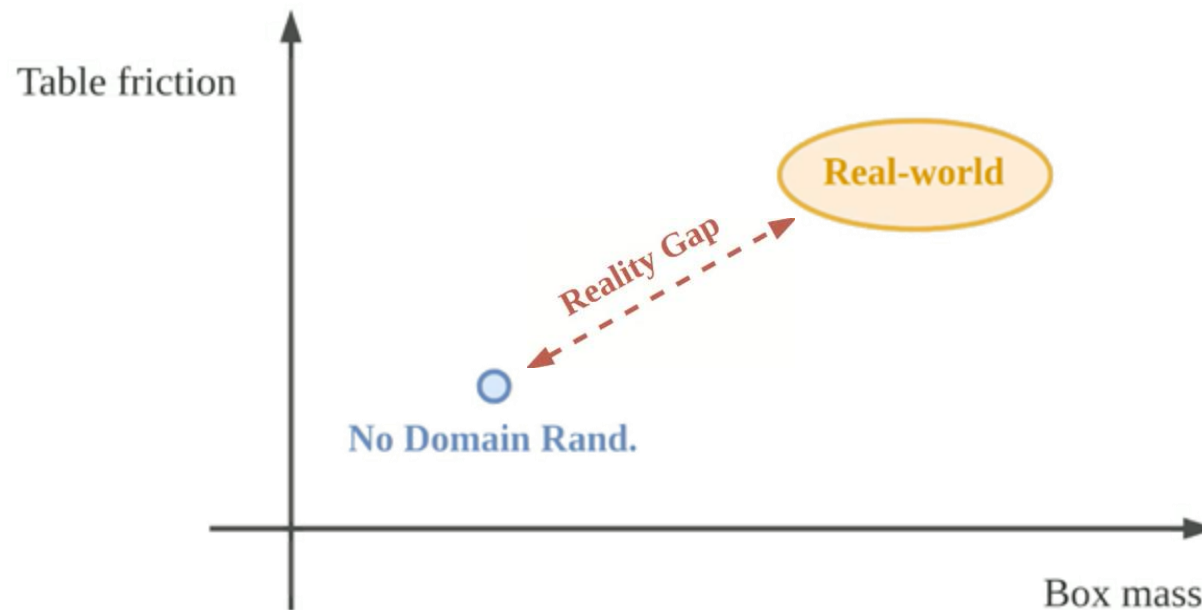
Testing in the real world



Solution:  
**Training with  
Domain Randomization**

# Sim-to-Real Transfer and Domain Randomization

**Problem:** teach a robot how to push a box to a target location



**Solution:**  
**Training with**  
**Domain Randomization**

# Sim-to-Real Transfer and Domain Randomization

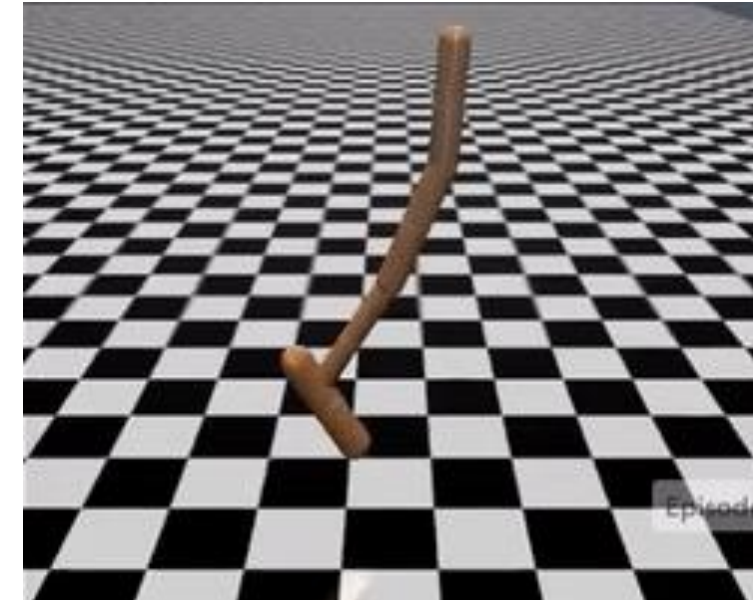
**Problem:** teach a robot how to push a box to a target location



**Solution:**  
**Training with**  
**Domain Randomization**

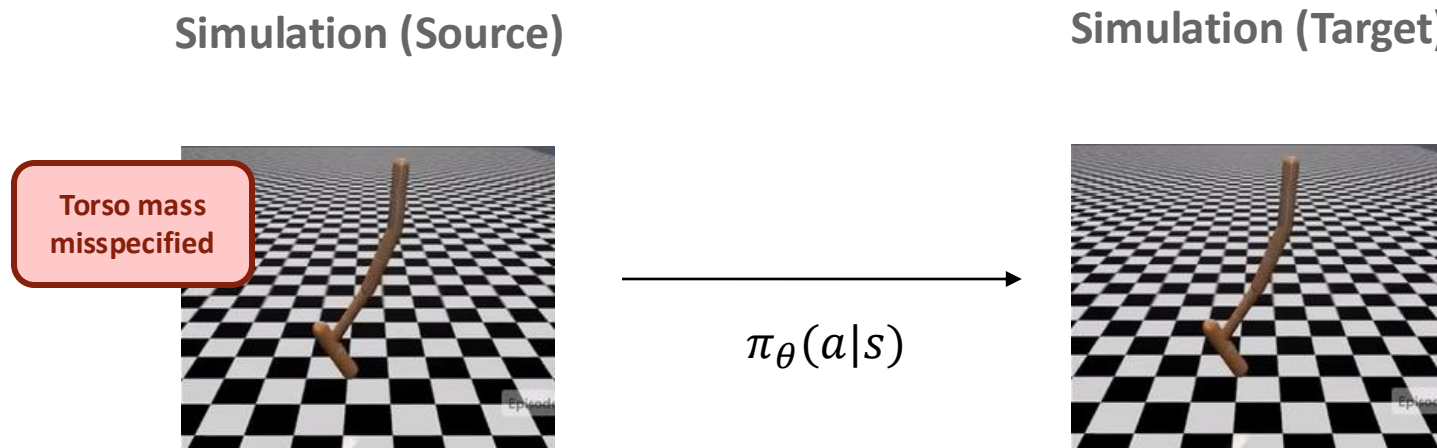
## Hopper environment

- Learn how to hop forward with a one-legged robot without falling, while achieving the highest possible horizontal speed.
- **Mujoco**
  - Physics engine for detailed, efficient rigid body simulations with contacts
  - Cross-platform GUI with interactive 3D visualization in OpenGL



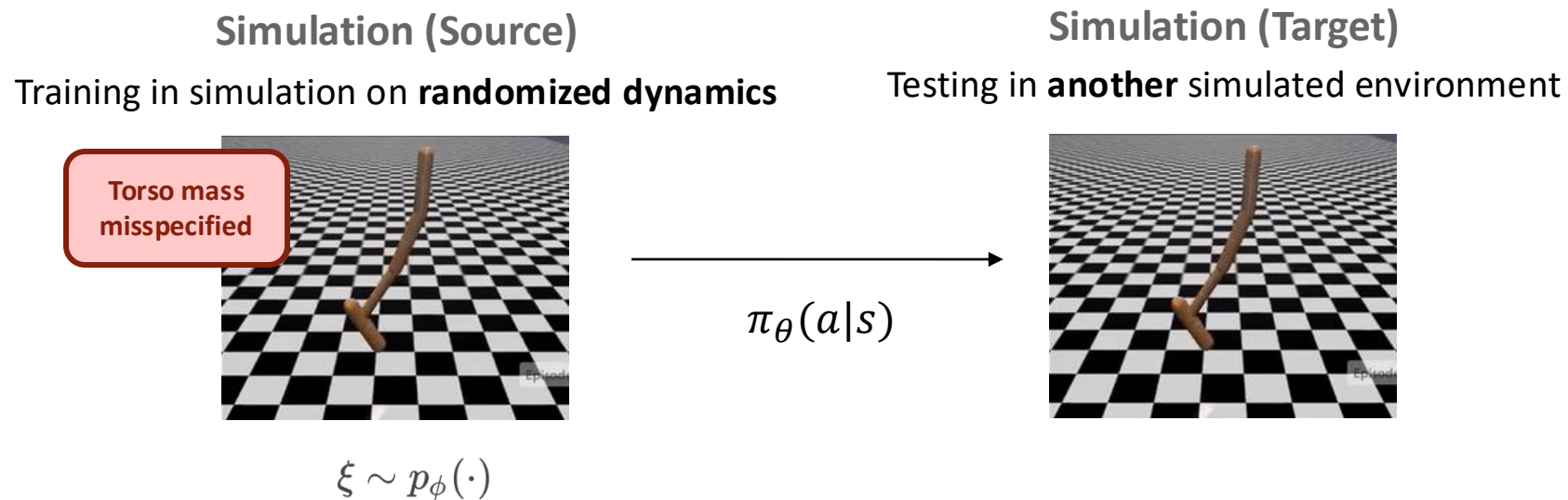
## Core part: Sim-to-Sim transfer

1. Train the Hopper agent with one algorithm of choice between PPO and SAC



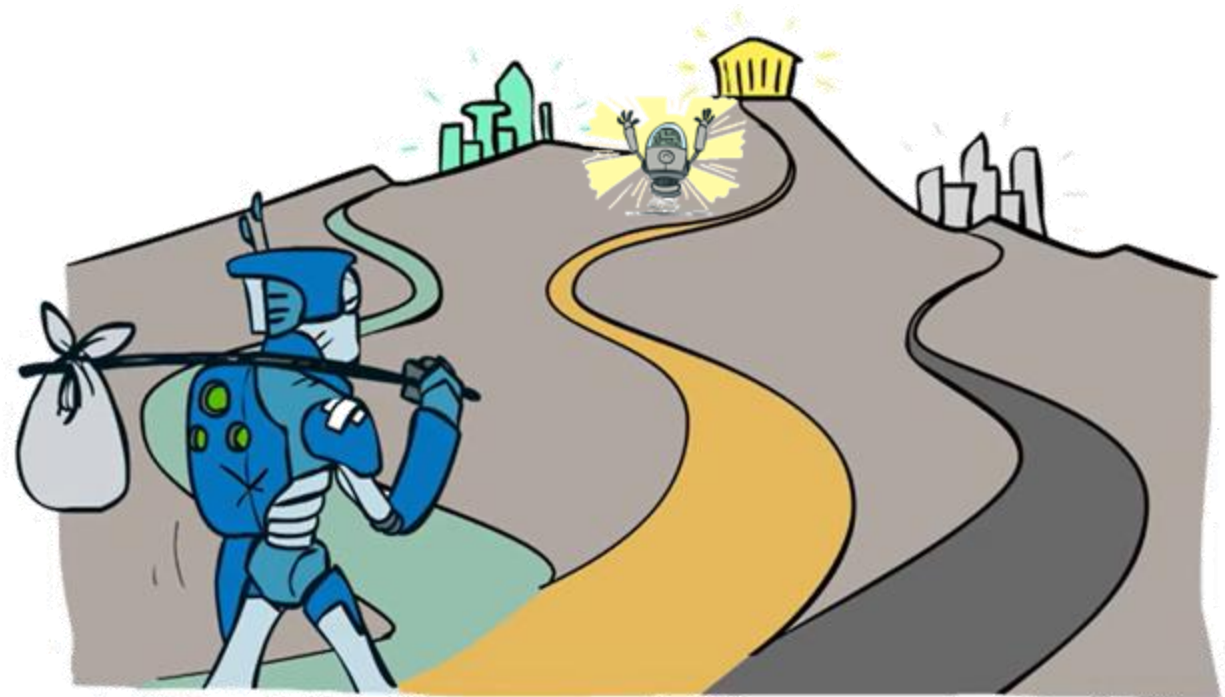
## Core part: Sim-to-Sim transfer

1. Train the Hopper agent with one algorithm of choice between PPO and SAC
2. Implement Uniform Domain Randomization (UDR) for the link masses of the Hopper robot



## Project extension

- It's your turn!
- Feel free to be ambitious
- Let your work reflect a research-inspired approach



# Mujoco Installation and other libraries

- Use conda environments

```
# pip install <lib>
```

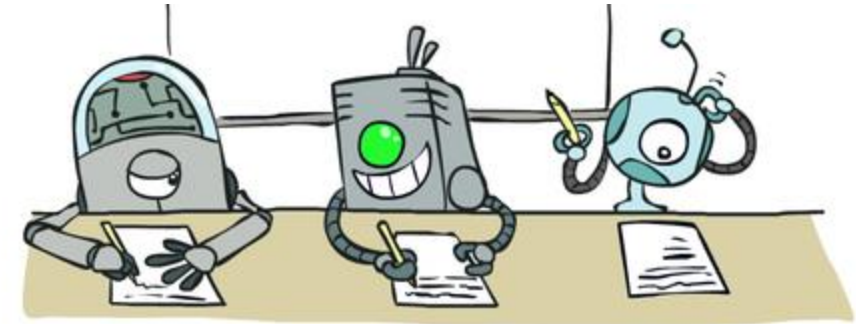
```
gymnasium
```

```
mujoco
```

```
stable-baselines3[extra]>=2.0.0
```



# Project Exam



- **Report and code:**
  - Core part (1.5 points)
  - Project extension (4 points)
  - Follow [this template](#) for the report
  - **Individual** submission for the core part (on the [personal repo](#))
    - **Deadline:** January 8th, at 23:59.
  - **Group** submission for the project extension (on the [team's repo](#))
    - **Groups are allowed** (up to 4 people)
    - **Deadline:** one week before the exam call
- **Oral exam** (20 points)
  - **Team members must take the exam on the same day**
  - Project presentation with slides (approx. 15 mins in total, divided equally between team members)
  - Theory questions